

Evaluation Report & Next-Step Execution Roadmap

Intern Name: SAGAR R

Assigned Construct: Quantitative Ability (Gq)

Proposed Task: Adaptive Quantitative Decision Arena

Evaluation Status: Proceed with Modifications and Enhanced Construct Purity

1. Executive Summary

The proposed Adaptive Quantitative Decision Arena demonstrates strong alignment with our vision of developing an AI-driven cognitive assessment platform that measures reasoning and quantitative competence through interactive experiences. The task incorporates pattern recognition, data interpretation, budgeting scenarios, and adaptive questioning mechanisms, making it highly engaging and suitable for personalized assessment. However, some scenarios currently extend beyond Quantitative Ability (Gq) and begin to measure Financial Literacy, Executive Functioning, and decision-making skills. To align with our startup objectives, the task should prioritize quantitative concepts, arithmetic reasoning, and numerical problem solving while using real-world scenarios only as contextual support.

2. Alignment with MentiScope's Objectives

- Alignment with CHC-Gq Theory: 8.5/10
- Scientific Validity: 8/10
- Innovation: 8.5/10
- Adaptive Potential: 9/10
- AI Readiness: 9/10
- MVP Feasibility: 9/10
- Overall Evaluation: 8.5/10 – Proceed with Modifications

3. Scientific Assessment

Primary quantitative components to measure:

- Arithmetic Reasoning
- Quantitative Concepts
- Data Interpretation
- Numerical Problem Solving
- Quantitative Comparisons

Potential construct contamination:

- Financial Literacy

- Executive Functioning
- Decision-Making Skills
- Reading Comprehension and Contextual Knowledge

The task should primarily assess quantitative reasoning processes rather than financial expertise or strategic planning abilities.

4. Recommended Task Architecture

Module 1 – Number Series and Pattern Recognition

Students identify numerical relationships and predict missing values.

Measures: Pattern recognition and arithmetic reasoning.

Module 2 – Quantitative Comparison

Students compare numerical quantities and determine equivalence or superiority.

Measures: Quantitative concepts and numerical reasoning.

Module 3 – Data Interpretation

Students analyze tables, graphs, and charts to answer quantitative questions.

Measures: Data interpretation and applied mathematics.

Module 4 – Applied Scenario

Students solve contextual problems involving budgeting and resource allocation.

Measures: Numerical problem solving while minimizing excessive dependence on financial literacy.

5. Recommended Scoring Model

- Arithmetic Reasoning Score – 30%
- Quantitative Comparison Score – 20%
- Data Interpretation Score – 25%
- Applied Problem Solving Score – 25%

6. AI Analytics to Capture

- Question accuracy
- Response time
- Error patterns
- Difficulty progression
- Hint dependency
- Strategy shifts

- Persistence score
- Data interpretation efficiency
- Adaptive learning curve
- Numerical reasoning profile

7. MVP Deliverables (Current Phase)

- ✓ 300-item question bank
- ✓ Difficulty calibration framework
- ✓ Adaptive routing engine
- ✓ Item exposure control mechanism
- ✓ Event logger
- ✓ Analytics dashboard
- ✓ Documentation
- ✓ Pilot dataset

8. Assessment SDK Compliance Requirements

All outputs must follow the common Assessment SDK specification.

Session Data:

student_id, session_id, construct, task_id, module_version, difficulty_level, start_time, end_time, timestamp.

Event Data:

event_id, student_id, session_id, construct, task_id, item_id, event_type, response, correct, reaction_time_ms, error_type, difficulty_level, timestamp.

Score Output:

raw_score, normalized_score, percentile, sub_scores, confidence_score.

Analytics Output:

behavioral_metrics and recommendations.

9. Next Sprint Execution Plan (2 Weeks)

Sprint Objective:

Refine the task architecture to improve construct purity and develop the foundational MVP prototype.

Expected Deliverables:

1. Literature review on Quantitative Ability (Gq) and CHC theory.
2. Construct map defining arithmetic reasoning, quantitative concepts, data interpretation, and numerical problem solving.
3. Design specifications for all four modules.
4. 300-item question bank with difficulty labels.
5. Adaptive routing engine design.
6. Difficulty calibration framework.
7. Event logging implementation.
8. Demonstration prototype of one adaptive module.

10. Definition of Done (DoD)

Scientific:

- ✓ Quantitative components clearly defined.
- ✓ Construct contamination minimized.
- ✓ Difficulty levels calibrated.

Technical:

- ✓ Item bank developed.
- ✓ Adaptive routing engine functional.
- ✓ Event logging and analytics dashboard available.
- ✓ Results exported in SDK format.

Validation:

- ✓ Internal consistency ≥ 0.80 .
- ✓ Pilot study with at least 100 students.
- ✓ Preliminary evidence of quantitative ability measurement.
- ✓ Adaptive engine performance evaluated.

Guidance to the Intern

Your responsibility is not to build the entire platform at this stage. Your responsibility is to build a scientifically valid micro-assessment engine for Quantitative Ability that follows the common Assessment SDK. Focus on creating adaptive quantitative tasks, calibrated item difficulty, and rich behavioral analytics. This module will later be integrated into the unified cognitive and career assessment platform.